NITTE (Deemed to be University) | NMAM INSTITUTE OF TECHNOLOGY

# A Project report on
# Sequence Generator and Evaluator using PSoC


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# CERTIFICATE

This is to certify that Abhiram, NNM23EC002 Anurag Shetty, NNM23EC022 Anvesh S Shetty, NNM23EC023 Deekshith Rajesh Rao, NNM23EC052 Bonafide students of N.M.A.M. Institute of Technology, Nitte have submitted a project report entitled "**Sequence Generator and Evaluator using PSoC**" as part of the Project based Embedded C using PSoC Laboratory, in partial fulfillment of the requirements for the award of Bachelor of Technology Degree in Electronics and Communication Engineering during the year 2025-2026.

**Name of the Examiner** **Signature with date**

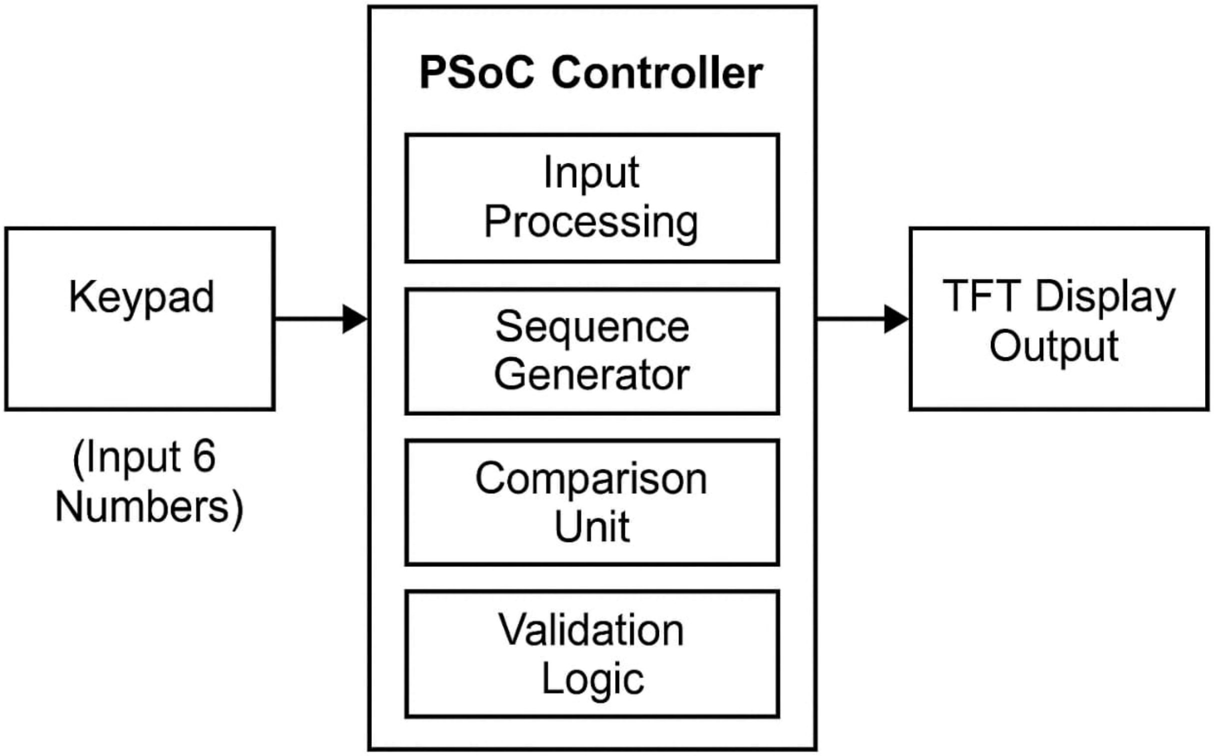
## Introduction:

This project implements an embedded sequence validation system using a TFT LCD display and a 4x4 matrix keypad. The user enters six numbers using the keypad, and the TFT LCD displays prompts, inputs, and validation results. The recurrence relation $P(n) = P(n-2) + P(n-3)$ is used to compute and validate the sequence.

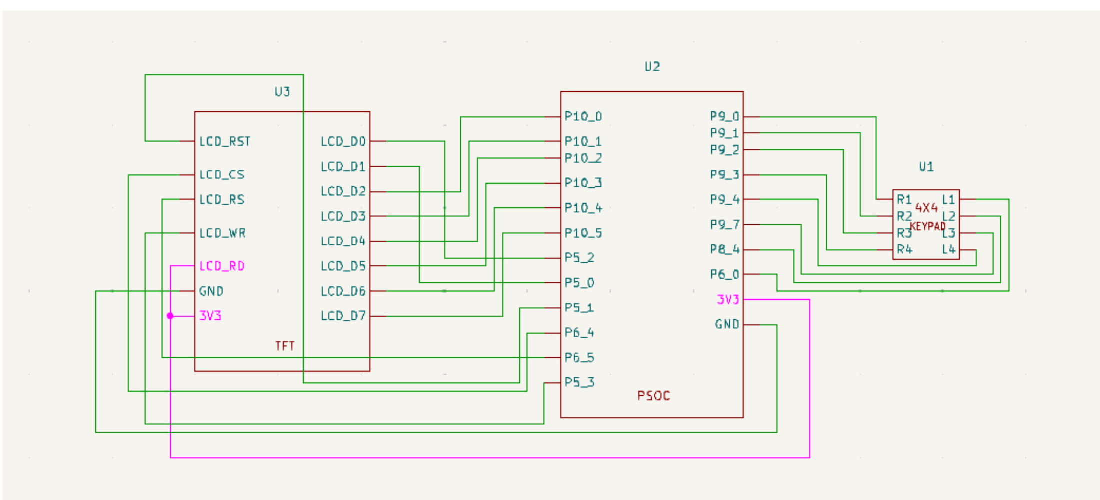
## Problem Statement:

The project requires the user to enter six numerical values using a 4×4 matrix keypad. When the system receives these values, it treats the first three numbers as the initial base sequence (X, Y, Z). Using these base numbers, the microcontroller computes the next set of three expected numbers using the recurrence relation $\mathbf{P(n) = P(n-2) + P(n-3).}$ Once the expected values are generated, they are compared with the last three numbers entered by the user. If the two sets match, the system declares the entire sequence valid. If the values do not match, the system classifies the sequence as invalid and highlights the error on the TFT LCD screen. This eliminates the dependence on a computer-based serial interface and instead uses a standalone keypad-based input mechanism.

## Block Diagram




## Schematic Diagram



## Objectives:

The primary objective of this project is to design a PSoC-based system that validates a sequence of numbers received through a 4x4 matrix keypad for user input. The system should process the six numerical inputs, apply the recurrence formula **$P(n) = P(n-2) + P(n-3)$,** and determine whether the user's sequence is valid. This includes capturing all user inputs through a keypad and displaying all system outputs through a TFT display.

## Implementation:

The firmware consists of three major components: the 4x4 matrix keypad input handler, the sequence computation module, and the TFT LCD for output display. The 4x4 matrix keypad handler reads user input as ASCII text, converts it into integers, and validates the formatting. The computation module processes the base values using integer arithmetic to generate the expected next three values. The comparison module checks the correctness of the input sequence. The TFT LCD module displays short messages such as "Valid Sequence" or "Invalid Sequence (–1)" using the SSD1306 I²C protocol. All functions execute within an infinite loop, enabling repeated testing without restarting the hardware.

1. Keypad Module:
   - ➢ Scans keyboard rows/columns.
   - ➢ Captures numeric input (0–9).
2. Computation Module:
   - ➢ Computes $P(n) = P(n-2) + P(n-3)$.

3. Display Module:

- Uses TFT LCD (SPI) to show prompts and results.

## System Requirements:

• PSoC microcontroller (Optiga Board)

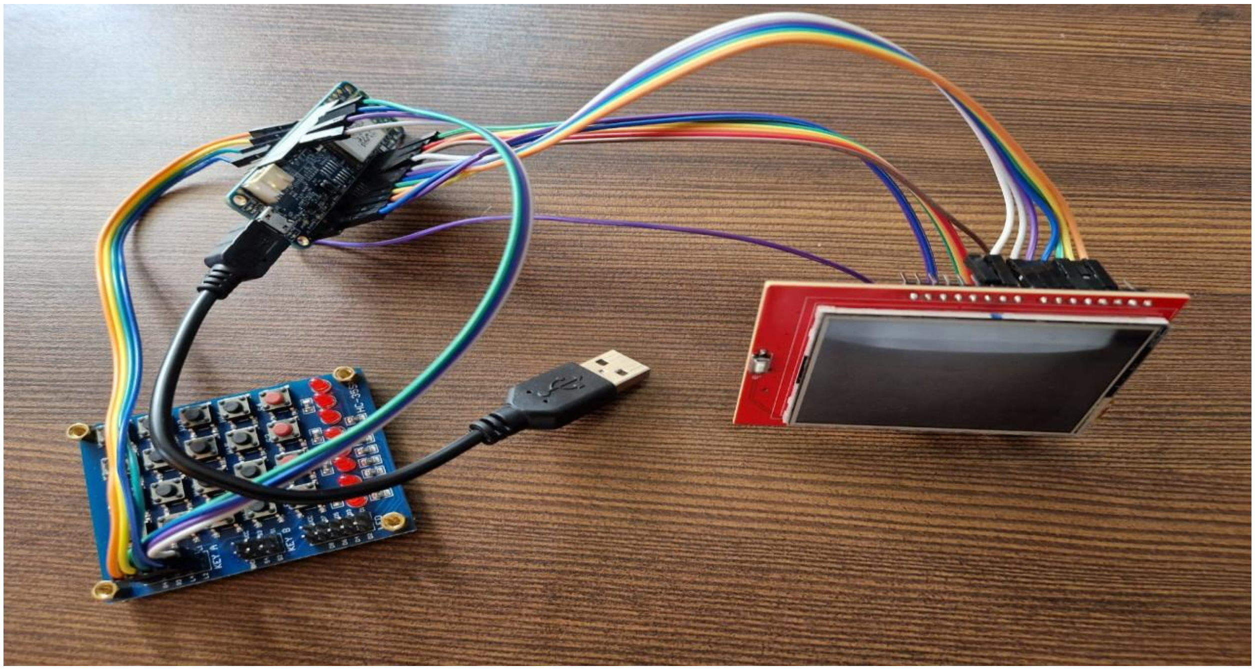
• 4x4 Matrix Keypad

• TFT LCD (ILI9341)

• Power supply & jumper wires

The hardware required for this project includes a PSoC microcontroller board, a 4×4 keypad for input, and a 3.2-inch TFT display with an ILI9341

breadboard, and a USB cable for power and programming. The software requirements include PSoC Creator or ModusToolbox, along with Embedded C code for keypad handling, sequence computation, comparison logic, and TFT display control.



## Methodology:

The project begins by initializing the PSoC, the keypad GPIO pins, and the TFT display. The TFT first shows a welcome or instruction message Sequence Generator. The system asks the user to add the number of digits in input sequence minimum three and the ask the number of digits to be present in the output sequence, which should be greater than input sequence, Then we have to enter the input sequence on the keypad. Each key press is debounced and stored. After receiving all input numbers, the system uses the input numbers to calculate the next numbers values using the recurrence rule. These computed values are then compared with the last three inputs entered by the user. If the values match, the TFT shows "Valid Sequence," otherwise it shows "Invalid Sequence." The screen also displays the six input values and the three computed output values for user verification.

## Output description:

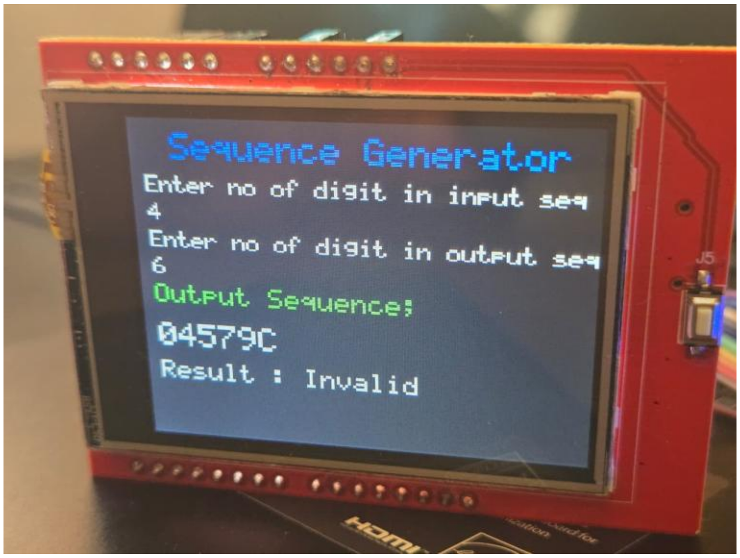
The TFT display shows the complete interaction with the system. First, it displays the six numbers entered by the user. Then it shows the three values generated using the recurrence formula. Finally, it prints the validation result, indicating whether the sequence is valid or invalid. This ensures that the user can clearly see the input, the generated output sequence, and the correctness of the result.

## Valid sequence Output:

If the user enters a correct sequence such as 0 1 2 1 3 3, the system computes the next three values as 1, 3, and 3. Since these values match the last three user inputs, the TFT displays “Valid” along with the generated output. This confirms that the user followed the correct recurrence rule


## Invalid Sequence Output :

If the last three numbers entered by the user do not match the computed values, the TFT displays “Invalid “This indicates that the user’s prediction is incorrect and the sequence does not follow the required recurrence rule.


Sequence Generator
Enter no of digit in input seq
4
Enter no of digit in output seq
6
Output Sequence:
04579C
Result : Invalid